

RBTM: Prediction of COVID-19 from Random Blood Test with Machine Learning Algorithms

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Abstract—Recently, the ongoing global pandemic of novel coronavirus infection had a devastating impact worldwide. We develop an efficient classification model that effectively produces the predictive values of infected patients with suspicious symptoms and epidemiological history to defeat this. The research aims to use the Traditional technique to compare clinical blood tests of positive and negative cases. The diagnostic Machine Learning model incorporates 551 random blood samples with the following parameters of the patient's demographic features, Platelet, Hemoglobin, Lymphocyte, Neutrophil, Leukocyte (WBC), Turbidimetric, Troponin-I of COVID positive and negative cases. The prediction model can achieve the classification report of Accuracy, Precision, Recall, and F1 score values. In this analysis, considering seven different algorithms for the prediction and the observation's estimation, the data is 5-fold cross-validated. Finally, investigational outcomes attain accurate predictions. Logistic Regression predicted 0.83% of accuracy. The Receiver Operator Characteristic (ROC) metrics for Logistic Regression, the Precision was 0.78%, Recall was 0.85%, and F1-score was 0.82%, Specificity was 0.58%, and Sensitivity was 0.41%.

Keywords: COVID-19, Coronavirus Variants, Machine Learning Techniques, Blood Test, ROC Metrics, Visualization.

I. INTRODUCTION

Coronaviruses are essential human pathogenies. The newly encountered deadly virus in Wuhan is accountable for collecting pneumonia cases. Rapidly it increases the spread worldwide quickly and leads to a comprehensive epidemic [1]. The origin of the infectious virus family is massive that causes the common cold mild-to-moderate ailments [2]. The impact leads to a drastic change in human life, and over 6.5 million are already affected all over the nations, and the fatality rate is confirmed over 5.3 million [3]. The wide ranges of clinical manifested symptoms are mild to moderate. The other signs that are considered respiratory are high-temperature fever, persistent cough, breathing issues, and sore throat [4]. ARDS, respiratory sepsis, and septic shock are included [5]. Immediate medical attention is required unless symptoms in shortness of breath, pressure in the chest, and changes in the countenance [6].

The coronavirus is associated with severe hypoxia, and the level of risk is high. It may lead to deterioration, multiple organ failures, and death. The essential elements of case definitions are: suspected, probable, and confirmed [7].

A. COVID Classification defines the following Variants

Delta (B.1.617.2 lineage): The Delta variant is exceptionally transferable, more than Alpha (B.1.1.7), more contagious than the earlier SARS-CoV-2 heredity. Surveillance report from the UK states the number of contaminations brought by Delta amplified as Alpha reduced, with 13.6% of substandard domiciliary infections linked with Delta compared to 9.0% for Alpha [8].

Omicron (B.1.1.529 lineage): Botswana was the first country to report this new variant omicron, pursued by South Africa in November 2021. A more significant number of 30 mutations are in the spike protein, increasing transmissibility and decreasing susceptibility to neutralizing antibodies [9].

Alpha (B.1.1.7 lineage): In late 2020 in the UK, this variant was identified and increased regional infections. Like other strains, it contains more than a dozen mutations from those multiple mutations found in spike protein [10].

Beta (B.1.351 lineage): It is, also named 20H/501Y.V2, was noticed in South Africa in late 2020. It has several spike proteins and phylogenetically varies from B.1.1.7 [11]. The mutations in E484K affect immunity to prior infection or vaccination. Many studies have focused on the beta; post-vaccination plasma will not neutralize viral constructs [12].

Gamma (P.1 lineage): This variant, also designated as 20J/501Y.V3, was initially detected in Japan. Subsequently, 42% of 31 prioritized specimens were reported in the Brazilian state of Amazonas in December 2020 [13]. The study shows that three mutations in the receptor-binding domain of the spike protein, namely N501Y, E484K, and K417T increased transmissibility and developments in immunity [14].

This work considered blood samples from COVID patients for exploration. Instantaneous processing and

elevated reliability involved multifarious approaches for interpreting the disease [15].

The rest of the manuscript formed as follows: Section 1 presented an introduction and previous corresponding works regarding state-of-the-art methods and varied clinical parameters utilized. The healthcare samples collected and the techniques used are described next. The resulting analysis of the comparison and ROC interpretation metrics is then delivered. A concise discussion and conclusion are provided in the final section.

II. MATERIALS AND METHODS

A. Related Work

This corresponding work examines several state-of-the-art techniques and the use of clinical data to determine the most appropriate analysis. (Samin Babaei Rikan et al. 2022) The author Applied a four-fold cross-validation method to determine the best approach. Identifying patients with SARS-CoV-2 infections can be done via random blood tests. Evaluated these models through paired samples t-tests for statistically significant differences [16].

(Bruno C'esar Feltes et al. 2022) Based on CBC data and suspected SARS-CoV-2 condition completed the analysis. They utilized four well-performing categories of feature selection Filter, Wrapper, Embedded, and Hybrid. The most considered attributes are red blood cell count, neutrophils and lymphocytes, eosinophils, monocytes [17].

(Isaac Nunez et al. 2021) The author carried out hematology tests and biochemical identifications to identify the critical illness—considered platelets, lymphocytes clinical parameters. Based on $P < 0.001$ significant, looking at the risk of death, hospital admissions for critically ill patients [18].

B. Data Collection

We have collected the dataset for this study from open sources to forecast comparative analysis. There are 551 cases in the original .csv file, each with 26 attributes. Patients in the study have conducted specific tests and other laboratory procedures. Handled some missing values through preprocessing: First, removed the features with the most missing values, leaving 26 attributes, including the blood count. After that, to avoid inconsistency, cleared samples with null values. Negative and positive instances are balanced and re-entered values for further analysis.

TABLE 1: CLINICAL DEMOGRAPHICS AND LABORATORY CHARACTERISTICS FEATURE AND DATA TYPES

Features	Data Type
Patient No	Numerical
Age	Numerical
Gender	Numerical
Platelet Distribution Width (PDW)	Numerical
Mean Platelet Volume (MPV)	Numerical

Platelet (PLT)	Numerical
Erythrocyte Distribution Width (RDW)	Numerical
Average Erythrocyte Volume (MCV)	Numerical
Hemoglobin (HGB)	Numerical
Lymphocyte Count	Numerical
Neutrophil Count	Numerical
Leukocyte (WBC)	Numerical
CRP (Turbidimetric)	Numerical
LDH	Numerical
Creatine Kinase (CK) (Serum/Plazma)	Numerical
Creatine kinase-MB (CK-MB)	Numerical
Troponin I	Numerical
Kreatinin	Numerical
AST	Numerical
ALT	Numerical
D-dimer, quantitative	Numerical
Procalcitonin	Numerical
PT (INR)	Numerical
PT (sn)	Numerical
APTT	Numerical
class	Numerical

The diagnostic information depends on clinical numerical data. Table 1 shows the extracted Clinical demographics and laboratory characteristics features and data types. All features are binary labels; negative (0) result represents not affected by the emerging disease, and positive (1) impact means confirmed.

C. Data Preprocessing

Preprocessing is cleaning to select the best components. The following are the steps [19]:

Cleaning: Process of removing variables with missing values.

Encoding: Associating the qualitative significances of the value.

Imputation: Removing missing values or replacing them for future operations.

Standardization: Placing segments and target variables.

Features selection: Features selection: Determining the best ten feature variables using statistical methods, and Figure 1. Feature Importance of Tree-Based Classifier.

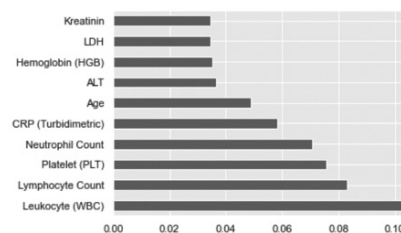


Fig. 1: Representation of Feature Importance of Tree-Based Classifier

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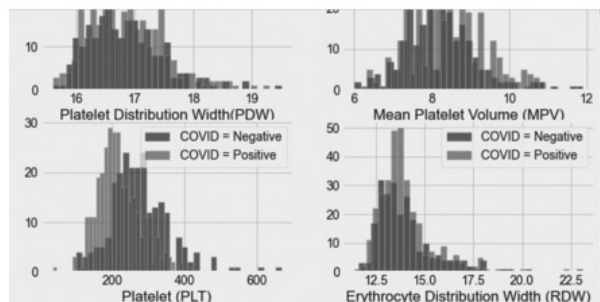


Fig. 5: Representation of Covid Positive, Negative Count

Figure 5. shows the Covid positive and negative count based on the attribute values.

B. Comparative Analysis

We split the dataset into two parts: training and testing, using 80% of the data for training and 20% for testing. We had 440 sample results for training, while for testing, we had 111. During model training, we selected the parameters that achieved the highest accuracy. A model evaluation factored in the accuracy of test data.

A comprehensive investigation of these classification techniques is being discussed, and both the experimental and visualization results are presented. The approach of comparative analysis is displayed in Table 2. The framework is initiated with the data acquisition step, then followed by feature extraction. The extracted features were analyzed to determine correlation coefficients. Then features are passed to the final classification. The classification results are given, and observations and improvements are made. The comparative analysis method achieved a classification accuracy of 0.83 percent for the given blood test data set of 551 patients with 26 attributes. The ROC classification report was also provided. Table 2 shows the comparative analysis of the predicted algorithms.

TABLE 2: COMPARATIVE ANALYSIS OF THE PREDICTED ALGORITHMS

Classification Algorithm	Accuracy Achieved
Logistic Regression	0.83%
GaussianNB	0.78%
Random Forest	0.76%
SGDClassifier	0.72%
Decision Tree	0.70%
K-Nearest Neighbor	0.67%
Support Vector Classification	0.64%

Fig. 6. shows the classification comparison in Box plot representation

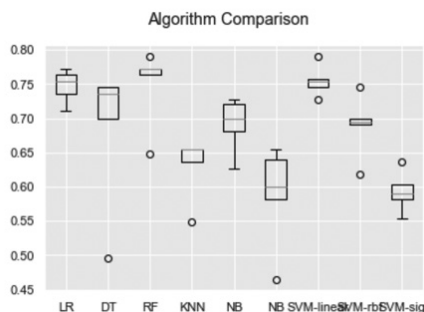


Fig. 6: Box plot for the Representation of Algorithm Comparison

Here is a summary of the test performance. Assess the accuracy performed the ROC analysis (Table 3).

TABLE 3: RECEIVER OPERATING CHARACTERISTICS ANALYSIS

Classifier	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.77	0.78	0.85	0.82
GaussianNB	0.58	0.53	0.58	0.54
Random Forest	0.77	0.76	0.77	0.76
SGDClassifier	0.57	0.32	0.57	0.41
Decision Tree	0.59	0.60	0.59	0.60
K-Nearest Neighbor	0.57	0.57	0.57	0.57
Support Vector Classification	0.66	0.43	0.66	0.52

IV. DISCUSSION AND CONCLUSIONS

Random blood tests have become more accessible worldwide. Predicting positive cases of COVID-19 through random blood tests makes it possible to prevent its spread. The disease's mortality and the possibility of producing new virus mutations can be limited. The result analysis shows that the comparative analysis has a reasonable level of accuracy for the blood test dataset. Symptoms, lymphocyte count, and disease severity determined the specific response to SARS-CoV-2. Whole blood tests showed good correlation values. In whole blood tests, the results of the classifier show the accuracy of the SARS-CoV-2 positive/negative classification. The Logistic Regression classifier method had the highest accuracy of 0.83%, Precision was 0.78%, Recall was 0.85%, F1-score was 0.82%, Specificity was 0.58%, and Sensitivity was 0.41%. The limitation of this work is the low number of patients studied within an observational study, which prevented a full evaluation of the potential of this prediction approach for supporting the diagnosis. Future work on improving the performance of the hybrid machine learning model will include many datasets. To improve overall accuracy and speed up computation, use clinical and blood test data that contains other variants of COVID-19.

Dataset Availability

Obtained the COVIDXCELL from the author Zerrin Gamsızkan (2021)

https://figshare.com/articles/dataset/Untitled_Item/14578692?file=27980028.

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